

Project Initialization and Planning Phase

Date	28 June 2026
Team ID	
Project Title	Opti Crop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed solution, **OptiCrop**, is an AI-powered smart crop recommendation system that assists farmers in selecting the most suitable crop based on soil nutrients (N, P, K), temperature, humidity, pH, and rainfall. By leveraging machine learning algorithms, the system provides accurate crop recommendations, helping farmers improve productivity, reduce crop failure, and promote sustainable agriculture.

Project Overview	
Objective	To develop an AI-based crop recommendation system that predicts the most suitable crop using soil nutrients and environmental conditions, enabling farmers to make informed farming decisions.
Scope	The project focuses on crop prediction using machine learning, data analysis, and a user-friendly web interface. It supports sustainable farming and can be extended with weather forecasting and fertilizer recommendations.
Problem Statement	
Description	Farmers often select crops based on experience rather than scientific analysis, leading to poor yields and financial losses due to changing soil and weather conditions.
Impact	Incorrect crop selection results in reduced productivity, unnecessary fertilizer usage, financial losses, and inefficient utilization of agricultural land.
Proposed Solution	
Approach	Collect soil and environmental parameters, preprocess the data, train a machine learning model, and provide real-time crop recommendations through a simple web application.
Key Features	- AI-based crop recommendation using Machine Learning. - Prediction based on N, P, K, temperature, humidity, pH, and rainfall.

Resource Requirements

Resource Type	Descriptio	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of	T4 GPU
Memory	RAM	16 GB
Storage	Disk space for data, models, and logs	512 GB
Software		
Frameworks	Python	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	ID	Jupyter Notebook, pycharm
Data		
Data	source,size, format	Kaggle data set, crop recommendation dataset